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The real campaign: The role of authenticity in the effectiveness of advertising disclaimers in digitally enhanced images

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ABSTRACT

In a series of three experimental studies, this research examines the effects of a retouch-free vs. retouched disclaimer on appearance satisfaction, advertising effectiveness, and evaluation of other women. Our results show the positive effect of a retouch-free disclaimer on appearance satisfaction, attitude toward the ad, and purchase intentions (Study 1) based on authenticity and its components: realism and meaningfulness of the ad (Study 2). Interestingly, results are in the opposite direction considering the evaluation of other women, who are evaluated as less attractive when exposed to a retouch-free disclaimer (Study 3). This study provides valuable insights from a theoretical and managerial perspective on how marketers can appeal to Millennials without overtly telling them they are authentic.

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1. Introduction

Digitally altered images have characterized the advertising industry throughout the past two decades. They have been considered acceptable as long as the images portrayed were not misleadingly exaggerating the effects of the product advertised. However, in light of the increase in eating disorders and lowered self-esteem facing younger generations, public policy makers, legislators, and academic researchers across the world have begun to question the negative effects of these images on self-satisfaction among young adults. For example, Israel in 2012 (Associated Press, 2012) and France in 2015 (Samuel, 2015) enacted laws that require a disclaimer for any digitally enhanced photographs. Several legislators are currently advocating similar regulations in the United States. Specifically, the "Truth in Advertising Act" of 2014 and 2016 calls for the Federal Trade Commission (FTC) to figure out how to handle potentially misleading altered images and their effects on consumers' wellbeing. The reason for this interest and plea for regulations is that these digitally enhanced advertising images often give people an unrealistic ideal of physical attractiveness.

Most of the media images consumers are exposed to have been digitally enhanced. Models depicted in ads are airbrushed to perfection. Skin color and tone are altered, imperfections are removed, and body shapes are modified. The result is an unachievable beauty ideal: an unrealistic image of female beauty impossible for the average woman to achieve. Recently, however, a countermovement has been noted in the advertising industry. For example, Aerie Real Campaign used disclaimers on their ads explicitly stating, "The girl in this photo has *not* been retouched" (e.g., see Krupnick, 2014). These retouch-free campaigns intend to illustrate that nobody is perfect, not even advertising models. In other words, this type of campaign aims to correct the false idea of attractiveness people get from exposure to retouched advertising campaigns and sheds a positive light on companies and products endorsing this message. Millennials are commonly identified as the target audience for these campaigns, as they are a leading force in the market by representing over 75 million consumers in the US (Byron, 2016) with a buying power of over $200 billion (Schwabel, 2015). Millennials are the new arbiters in the advertising and marketing industry. They dictate the rules in terms of innovations and trends, making them marketers' best allies when it comes to spreading advertising messages. However, traditional advertising seems to be rather ineffective on this group. Millennials are advertising and marketing literate, equipped with knowledge that makes them skeptical toward advertising. According to Schwabel (2015), Millennials trust people over logos and brands, and therefore authenticity becomes paramount. How do we appeal to Millennials without telling them we are indeed authentic? This study offers insights into this question by considering the context of a real campaign that employs a retouch-free disclaimer and its effects on appearance satisfaction, attitude toward the advertisement, and purchase intentions. This research extends existing literature on social comparison theory and media exposure (Tiggemann, Slater, Bury, Hawkins, & Firth, 2013; Tiggemann, Slater, & Smyth, 2014; Want, 2009; Slater,

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Tiggemann, Firth, & Hawkins, 2012; Grabe, Ward, & Hyde, 2008; Bury, Tiggemann, & Slater, 2014) as well as the role of authenticity in marketing (Beverland, Lindgreen, & Vink, 2008; Napoli, Dickinson, Beverland, & Farrelly, 2014; Rose & Wood, 2005) in order to explore conditions under which a real campaign might be effective.

Specifically, our paper offers three studies. Study 1 investigates whether depicting an advertising model followed by the explicit statement that she "has not been retouched" (retouch-free campaign) versus that she "has been retouched" (retouched campaign) yields more positive self-evaluations (appearance satisfaction) and advertising effects (attitude toward the ad and purchase intentions) compared to a control condition without a disclaimer. Study 2 tests the underlying mechanisms driving these effects in terms of authenticity and its components. Study 3 explores the effects of a real campaign on the evaluation of other (non-model) women.

2. Advertising disclaimers, appearance satisfaction, and advertising effectiveness

Advertising disclaimers are statements used to clarify potentially deceptive aspects of the ad, including statements offering advice and providing supplemental information to the viewer within the commercial. They have mainly been investigated in the context of advertising to children (Bakir, 2009) and in the context of food and drug advertising (Wicks, Warren, Fosu, & Wicks, 2009). Early consumer research has suggested that advertising might affect consumers when they compare themselves (explicitly or implicitly) with the idealized images and lifestyles portrayed in ads. Specifically, Richins (1991) explored the effects that exposure to advertising with highly attractive models has on female college students' self-perceptions of physical attractiveness. However, no significant effect was found to support the hypothesis that advertising images might lower their self-perception of physical attractiveness. The following section outlines the main findings since Richins (1991) and frames our hypotheses related to the effectiveness of a real campaign on appearance satisfaction and advertising effectiveness.

2.1. Media influence on appearance satisfaction

A plethora of correlational and experimental studies have been offered in support of the negative link between exposure to attractive models and the overall satisfaction women have with their bodies (Heinberg, Thompson, & Stormer, 1995; Grabe et al., 2008; Groesz, Levine, & Murnen, 2002; Want, 2009). Most of these studies have identified the sociocultural theory of body image (Thompson, Heinberg, Altabe, & Tantleff-Dunn, 1999) and the social comparison theory (Festinger, 1954) as the most robust theoretical accounts for understanding the negative relationship between exposure to ideal beauty and appearance satisfaction. According to the sociocultural theory, mass media creates and reinforces societal beauty ideals (Groesz et al., 2002) and the current cultural ideal of beauty seems to emphasize a young, tall, and thin woman (Slater et al., 2012). Social comparison theory (Festinger, 1954) focuses on how individuals evaluate "the self" by comparing themselves to others. The theory suggests that individuals engage in downward (comparing themselves to someone they perceive to be of less value) or upward (comparing themselves to someone they perceive to be of more value) comparisons as a way of judging themselves (self-evaluation) or increasing their judgment of the self (self-enhancement) (Wills, 1981). In fact, self-enhancement is one of the ultimate motives for self-evaluation (Wood, 1989). According to the theory, individuals are motivated to get a sense of validity (self-enhancement) by comparing themselves against a specific benchmark provided by other individuals. This benchmark might be an ability or trait in which the individual might exert some sort of control (i.e., appearance). In the context of advertising and exposure to media images, individuals—especially females—learn at an early age to form opinions about their own attractiveness compared to the attractiveness of models portrayed in ads by engaging in upward comparisons (Hargreaves & Tiggemann, 2004). How close or far women feel in terms of attractiveness to a model portrayed in the media is the basis of individuals' self-evaluation of attractiveness.

Grabe et al. (2008) suggest that media literacy prevention programs are effective in teaching individuals how to be active and critical consumers. Furthermore, some empirical studies (e.g., Ogden & Sherwood, 2008) support the possibility to reduce or minimize the negative effects of media on appearance satisfaction through media education. However, research is scant.

2.2. A retouched versus retouch-free disclaimer

The logic of the social comparison theory and the initial evidence provided by studies on the negative role of media on appearance satisfaction suggest that labeling images as digitally enhanced should reduce the negative effect of comparisons since individuals would be reminded of the unrealistic targets of their comparisons and would therefore experience more appearance satisfaction. Yet, few studies empirically support this hypothesis. Slater et al. (2012) experimentally investigated the role of a traditional disclaimer ("These images have been digitally altered") on digitally enhanced images used in print ads on women's appearance satisfaction and found significant results. However, these findings were not supported by two empirical studies run by Tiggemann et al. (2013).

Considering the current trend of some companies using non-digitally altered images of women and labeling these images explicitly as not digitally enhanced, we question the effect of a retouch-free disclaimer (where the ad disclaimer claims to use the model image without any retouch), versus a retouched disclaimer (where the ad disclaimer claims the model image has been retouched), campaign on women's appearance satisfaction. No research to our knowledge has compared a retouch-free disclaimer to a retouched disclaimer in regard to appearance satisfaction. One exception is offered by Tiggemann et al. (2014), who hypothesized that a retouch-free disclaimer (i.e., "This model has *not* been retouched") might actually give rise to greater upward social comparison and therefore less appearance satisfaction. They relied on work by Kulik and Gump (1997), who suggested that individuals seek out comparisons more with others who are similar on related attributes (i.e., normal women) than ideal ones (i.e., professional models). Therefore, from their perspective, the portrait of an attractive model and the claim that she has not been retouched might activate greater body dissatisfaction. However, this was not confirmed in their results. Tiggemann et al. (2014) found that body dissatisfaction increased after exposure to the thin ideal, but there was no significant effect of labels (retouch-free vs. retouched) on how women felt about their appearance. Contrary to Tiggemann et al. (2014), we hypothesize a real (i.e., "This model has *not* been retouched") campaign is able to activate a beneficial upward comparison in which women might rate their attractiveness closer to the model they compare themselves with. The driver for this positive upward comparison is self-enhancement, which according to social comparison theory occurs when individuals look for positive traits in themselves based on the best person they are trying to compare themselves with. Indeed, research has shown that in the case of upward social comparison (i.e., an average female Millennial versus a professional model), individuals like to believe they are part of the elite or superior, and make comparisons highlighting the similarities between themselves and the comparison group, ultimately leading to a boost in their own appearance satisfaction (Collins, 1995; Suls, Martin, & Wheeler, 2002). Accordingly, labeling images as not being digitally enhanced (i.e., the real campaign) may activate self-enhancement goals and therefore increase one's own appearance satisfaction due to a positive upward comparison. As such, we hypothesize a positive effect on appearance satisfaction when using a retouch-free disclaimer. Contrarily, a retouched disclaimer may not activate this self-enhancement goal among its audience because it admits that the

model is "unreal", and therefore the model is not considered a valid benchmark for comparison, creating no need to fulfill any self-enhancement goals. Hence, we expect the retouched condition to exert a less positive effect on appearance satisfaction than a retouch-free condition.

H1. Exposure to a retouch-free disclaimer generates greater appearance satisfaction than exposure to a retouched disclaimer.

Moreover, we expect the retouch-free disclaimer will also exert a positive effect on advertising effectiveness in terms of attitudes toward the ad and intentions to purchase the brand. These expectations are guided by research in creative advertising (Dahlén, Rosengren, Törn, & Öhman, 2008; Pieters, Wedel, & Batra, 2010; Smith, MacKenzie, Yang, Buchholz, & Darley, 2007), which often capitalizes on something relevant to the consumer (Smith et al., 2007). A relevant advertisement refers to an advertisement being meaningful to the consumer (Modig & Rosengren, 2014). Social issues are often identified as key ingredients to create relevant or meaningful marketing communications that will resonate with consumers and their values (Becker-Olsen, Cudmore, & Hill, 2006; Murray & Vogel, 1997). Considering Millennials, they are particularly sensitive to social causes and cause-related marketing (Smith, 2011). In fact, according to an article published by Josephson (2016), Millennials like to give and want to give more to nonprofit organizations. As reported in the article "93% of Millennials gave to nonprofit organizations in 2010, with 21 percent giving $1,000 or more during the course of the year (Josephson, 2016)." According to the same article, they like matching donations, they want to volunteer, they are willing to fundraise, and they are most likely to donate (84% of Millennials studied) when they fully trust the organization. Millennials value authenticity over innovation (Beattie & Fernley, 2014). They support authentic brands that act in line with their values (Napoli et al., 2014).

A retouch-free campaign suggests the need to rethink the current way we design advertising messages and signals the willingness of brands to contribute to a process that attempts to combat the known negative effects of exposure to unrealistic images in advertising. Lange, Rosengren, and Blom (2015) demonstrate a positive link between creative advertising and the perception of high quality brands and products, which translates into a more positive attitude and a higher intent to buy (Modig & Rosengren, 2014; Sasser & Koslow, 2008; Smith, Chen, & Yang, 2008; Smith et al., 2007). Research in cause-related marketing and corporate social responsibility (Becker-Olsen et al., 2006) has emphasized how a focus on causes and responsibility of brands/products toward society at large promotes positive attitudes (Brown & Dacin, 1997) and increased intent to buy from the brand (Murray & Vogel, 1997). Therefore, we hypothesize and test in Study 1 that ads focusing on being retouch-free yield a more positive attitude toward the ad and a higher intent to buy from the brand.

H2. Exposure to a retouch-free disclaimer generates a more positive attitude and willingness to buy from the brand than exposure to a retouched disclaimer.

3. Study 1

Study 1—preceded by a pretest—investigates whether a retouch-free advertising campaign is more effective than a retouched campaign on appearance satisfaction (H1), as well as on attitude toward the ad and purchasing intentions from the brand (H2).

3.1. Method

3.1.1. Pretest stimuli selection

In order to create our stimuli, five images sourced online depicting a young female model in lingerie were selected for a pretest. The featured models were discussed in a focus group composed of fifteen young female college students from a southwestern university in the US in their second year of their business degree program (ages 22–25; 12 Caucasians, 1 Latino, 1 African American, 1 Asian; all appeared to have a healthy BMI). The 15-min discussion focused on ethnicity, self-identification, attractiveness, familiarity, and credibility if the image stated that it was not retouched. Subjects also indicated their familiarity with a fictitious brand name: ANEA. The image selected was evaluated to be the best on self-identification, attractiveness, and credibility of not being retouched. None of the subjects were familiar with the images, the source of the images, or the brand name ANEA.

3.1.2. Design and stimuli

All three conditions used the same selected image and fictitious brand name (ANEA), but the advertising disclaimer was different. One advertising disclaimer stated: "The model in this ad has been retouched" (retouched condition), while the other advertising disclaimer stated: "The model in this ad has *NOT* been retouched" (retouch-free condition). Additionally, a control condition without an advertising disclaimer was included. Appendix A reports the images used for the retouch-free and retouched condition. The control condition used the same image with only the brand name ANEA included in the ad.

A second pretest with 126 female college students from a southwestern university in the US was conducted in order to make sure the three ads we chose based on pretest 1 were not perceived differently on quality ($p = 0.319$), believability ($p = 0.601$), informational value ($p = 0.09$), and professionalism ($p = 0.241$). Their advertising literacy (4 items, $\alpha = 0.699$; scale adapted from Bearden, Hardesty, & Rose, 2001) was also measured in order to make sure that our sample of Millennials is indeed advertising literate ($M = 5.28$, $SD = 0.84$ on a 7-point scale).

3.1.3. Respondents and procedures

A sample of 230 female Millennials ($M_{age} = 21.54$, $SD = 1.87$ l; the majority being Caucasian) from a large US university participated in a 3-level, single-factor (advertising disclaimer: retouch-free vs. retouch vs. no disclaimer) between-subjects online experiment in a controlled lab environment. Millennials are young consumers, born between approximately 1985 and 1999 (Pendergast, 2007). Female Millennials were selected for this study since past research suggested salience of disclaimers on appearance satisfaction for females only (Tiggemann et al., 2013; Tiggemann et al., 2014; Want, 2009; Slater et al., 2012; Grabe et al., 2008; Bury et al., 2014). The great majority of the sample indicated they were pleased with their appearances as measured prior to the experiment (median = 74.5 and mean = 68.21 on a 100-point scale, $SD = 21.38$), with high overall self-esteem (median = 3.4 and mean = 3.33 on a 4-point scale, $SD = 0.49$). The sample also indicated a relatively positive mood prior to the experiment (median = 59.88 and mean = 57.95 on a 100-point scale, $SD = 22.46$).

3.1.4. Procedure and measures

Participants were randomly exposed to one of the three conditions and completed an online questionnaire in a controlled lab environment. The questionnaire contained the measures for the dependent variables (post appearance satisfaction, attitude toward the ad, and purchase intentions), control variables (prior mood and self-esteem) and demographics (age). Finally, the respondents were debriefed and thanked for their participation.

Prior to the start of the experiment, respondents' positive mood was measured by a four-item zero-to-one-hundred-slider scale (i.e., *"How do you feel right now?" Happy; Confident; Hopeful; Optimistic*) ($\alpha = 0.88$). Respondents' self-esteem was measured by ten items (source) on a four-point scale (e.g., "*I take a positive attitude toward myself*") ($\alpha = 0.84$). Additionally, respondents' prior appearance satisfaction was measured using the Visual Analogical Scale (VAS) suggested by Tiggemann et al. (2014) by asking "*How do you feel about yourself?*", ranging from *"not at all satisfied with my appearance"*(0) to *"very satisfied with my appearance*" (100). This appearance satisfaction measure was

initially pretested with 126 females against a multi-item body satisfaction scale which was measured by means of 13 items on a five-point Likert scale (e.g., "I feel good about my body", "On the whole, I am satisfied with my body", "I respect my body", etc.) ($\alpha = 0.871$). There was a positive, moderately strong, and significant correlation between the two measures ($r = 0.622$, $p < 0.001$). Therefore we felt confident using this measure instead of a multi-item measure because of its convenience. This latter question was repeated after exposure to the experimental stimuli, as a dependent measure for appearance satisfaction. Respondents' attitudes toward the ad were measured by means of a single-item seven-point semantic differential scale (i.e., ranging from "*I do not like the ad at all*" to "*I like the ad very much*"). Purchase intentions were measured by a single-item seven-point semantic differential scale (i.e., ranging from "*I have no intent at all to buy from the brand (ANEA)*" to "*I definitely intend to buy from the brand (ANEA)*").

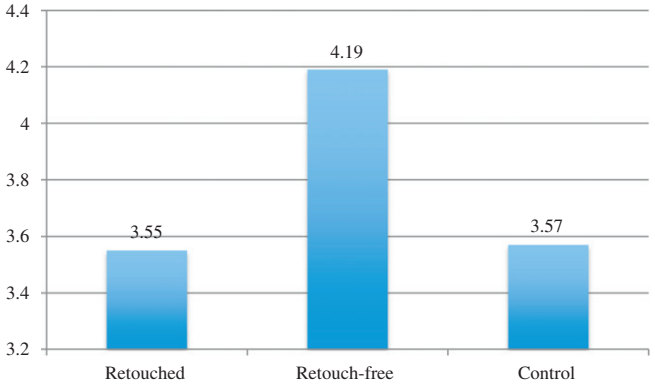


Fig. 2. Study 1 – attitude toward the ad results.

3.2. Results

3.2.1. Characteristics of the sample

We checked that the women in our sample ($N = 230$) did not differ on a number of core variables in each of the three experimental conditions. A series of one-way analyses of variance (ANOVAs) showed that women across the three experimental groups did not differ significantly on age ($p = 0.367$), prior mood ($p = 0.825$), and self-esteem ($p = 0.713$).

3.2.2. Results for appearance satisfaction

The results of this study showed a main effect of the three conditions on women's appearance satisfaction controlling for prior appearance satisfaction ($F(2, 229) = 3.117$, $p = 0.046$). Further t-testing showed that the retouch-free condition ($M = 58.94$) leads to a significantly higher appearance satisfaction than the retouched condition ($M = 48.82$) ($t(139) = -2.367$, $p = 0.019$) and control condition ($M = 49.42$) ($t(166) = 2.477$, $p = 0.014$), in line with H1. The retouched and control conditions do not significantly differ from each other ($t(149) = -0.134$, $p = 0.893$). Fig. 1 illustrates these results.

3.2.3. Results for advertising effectiveness

The results of this study have shown a main effect of the three conditions on women's attitudes toward the ad ($F(2, 227) = 5.83$, $p = 0.003$). Further t-testing has shown that the retouch-free condition ($M = 4.19$) leads to a significantly more positive attitude toward the ad than the retouched condition ($M = 3.55$) ($t(139) = -2.955$, $p = 0.004$) and control condition ($M = 3.57$) ($t(166) = 2.93$, $p = 0.004$) in line with H2. The retouched and control conditions did not significantly differ from each other ($t(149) = -0.113$, $p = 0.910$). Fig. 2 illustrates these results.

Additionally, we found a main effect of the three conditions on women's purchase intentions toward the brand ($F(2, 243) = 7.18$, $p = 0.001$). Further t-testing showed that the retouch-free condition ($M = 3.30$) yielded significantly higher purchase intentions than that of the retouched condition ($M = 2.75$) ($t(148) = -2.071$, $p = 0.040$) and control condition ($M = 2.40$) ($t(178) = 3.693$, $p = 0.000$), also in line with H2. The retouched and control conditions did not significantly differ from each other ($t(156) = -1.392$, $p = 0.166$). Fig. 3 illustrates these results.

3.3. Discussion

In Study 1 we found consistent main effects of the three conditions (retouched, retouch-free, and no disclaimer) on women's appearance satisfaction, attitude toward the ad, and purchase intentions. Women react more favorably to the condition in which the advertising disclaimer says the model has not been retouched than to both other conditions. This is in line with our reasoning that a retouch-free disclaimer activates motives of self-enhancement, because the retouch-free disclaimer basically states that the model is real and thus can be used as a benchmark for (positive) upward comparison. However, two limitations of Study 1 are that we did not test the realism of the model as a mediator nor did we control for possible alternative explanations as to why women react most favorably to the retouch-free disclaimer. For instance, the effect could also be attributed to novelty or the informational value of the ad. These alternative explanations will be ruled out in Study 2.

In Study 1 we found no significant difference in the effect of the retouched condition versus the control condition on women's appearance satisfaction, attitudes toward the ad, and purchase intentions. In other words, women react the same to an advertisement that comes with a "The model in this ad has been retouched" disclaimer, as to an advertisement with no disclaimer at all. We attribute this insignificant difference to Millennials' advertising literacy. Advertising literacy is

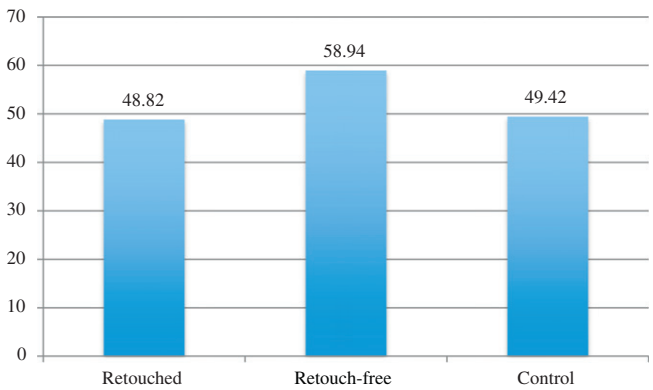


Fig. 1. Study 1 – appearance satisfaction results.

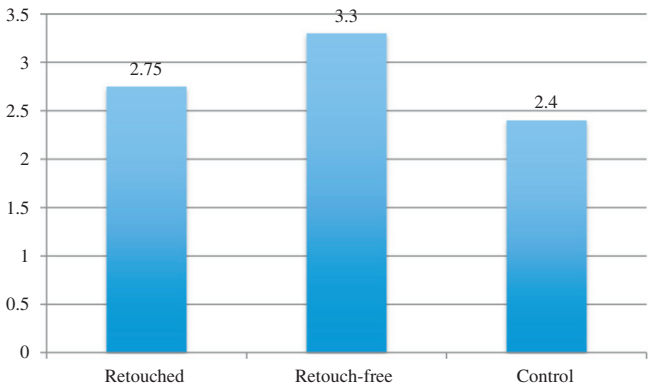


Fig. 3. Study 1 – purchase intentions of the brand results.

commonly defined as the ability to read and understand advertising content (Young, 1990). Literature on advertising literacy has suggested that the more advertising literate you are, the less susceptible you are to advertising effects (Wright, Friestad, & Boush, 2005). Young women in western societies generally possess relatively high advertising literacy due to their familiarity with persuasive media from an early age (Livingstone & Helsper, 2006). Hence, this could clarify the non-significant difference between the retouched and control condition in our study. Our pretest has shown that advertising literacy is moderately high among female Millennials. This finding has strong implications for the advertising industry. The legal measures some countries are taking to require the use of disclaimers on digitally enhanced images might be ineffective, as women are literate enough to understand that images are often tweaked. Labeling those images as "retouched" does not increase women's appearance satisfaction, attitudes, or purchase intentions compared to not labeling them. In other words, the advertising disclaimer is redundant.

Study 2 builds on Study 1 by focusing on the concepts of authenticity, realism, and meaningfulness as the underlying mediators driving the effects we found in Study 1. Additionally, Study 2 includes manipulation checks for the stimuli (novelty, informational value, attractiveness of the model, how current the ad is, and how substantive the ad is) to exclude potential alternative explanations for the effects we found.

4. The role of authenticity

H1 and H2 propose a positive effect of the retouch-free (vs. retouched) disclaimer on appearance satisfaction, attitude toward the ad, and purchase intentions. However, they do not explain why the effect occurs. Current advertising research highlights how authenticity is a value sought by consumers in processing advertising content (Beverland et al., 2008; Napoli et al., 2014; Rose & Wood, 2005). The concept of authenticity has been used to refer to sincerity, innocence, and originality (Fine, 2003), reality or truth (Kennick, 1985), and being natural, simple, unspun, and honest (Boyle, 2006). Authenticity is defined and experienced by consumers and influenced by their interest and knowledge of a specific subject (Grazian, 2005). A focus on authenticity is a must in contemporary marketing (Brown, Kozinets, & Sherry, 2003; Napoli et al., 2014). However, the use of advertising is believed to negatively affect the attempt to be authentic since consumers perceive methods of mass marketing to be far from authentic (Beverland & Luxton, 2005; Peñaloza, 2000; Peterson, 2005). Advertisers cannot simply state that their products, brands, or messages are authentic in order to have a positive effect. They need to be creative (Baack, Wilson, & Till, 2008) and convey key elements defining authenticity without making claims of being authentic. According to the aforementioned research, we believe that a retouch-free campaign is perceived to be more authentic, which in turn has a more positive effect on appearance satisfaction and advertising effectiveness. Based on previous research, we propose authenticity of the advertisement as the key explanation of the positive effect of a retouch-free disclaimer on our variables of interest. Hence, we propose that the authenticity of an ad mediates the positive effect of the retouch-free condition: because the retouch-free ad is perceived as more authentic, it exerts a positive effect on attitudes and purchase intentions.

H3. Authenticity of the ad mediates the effect of the retouch-free disclaimer on appearance satisfaction, attitude toward the ad, and purchase intentions from the brand.

From a practitioner standpoint, it is important to know which specific elements of authenticity might mediate the effect of a retouch-free disclaimer on appearance satisfaction. In other words: what key elements could practitioners use to frame their ads in order to make them more authentic? The study of authenticity in advertising has suggested that the construct relates to being real (Grayson & Martinec, 2004; Stern, 1994) and meaningful (Beverland et al., 2008). Studies on the role of media images on self-esteem have suggested that the realism of an image might exert a positive impact on body satisfaction (e.g., Tiggemann et al., 2013; Tiggemann et al., 2014; Want, 2009; Slater et al., 2012; Grabe et al., 2008; Bury et al., 2014). Meanwhile, research on cause-related marketing and corporate social responsibility has underlined how values, as they relate to being meaningful to the consumer, have a positive effect on attitude and intents to buy (e.g., Becker-Olsen et al., 2006; Brown & Dacin, 1997; Murray & Vogel, 1997). Therefore, we hypothesize that the realism of a model mediates the effect of the retouch-free disclaimer on women's appearance satisfaction, whereas meaningfulness of the ad mediates its effects on attitudes and purchase intentions. We test this hypothesis in Study 2.

H4. The effect of the retouch-free disclaimer on appearance satisfaction is mediated by the realism of the model in the advertisement.

H5. The effect of the retouch-free disclaimer on attitude toward the ad and purchase intention from the brand is mediated by the meaningfulness of the advertisement.

5. Study 2

Our second study tests the underlying reason why the retouch-free condition yields more optimal results, in terms of both appearance satisfaction and advertising effectiveness (i.e., attitudes toward the ad and purchase intentions), than the retouched condition or the control condition without a disclaimer. In particular, we propose that the authenticity of the ad and its components (i.e., realism of the model and meaningfulness of the ad) mediate the effects of our conditions on appearance satisfaction, attitudes toward the ad, and purchase intentions (H3, H4, and H5).

5.1. Method

5.1.1. Design and stimuli

Study 2 comprises a 3-level 1-factor (advertising disclaimer: "retouched" vs. "not retouched" versus control group containing no disclaimer) between-subjects online experiment in a controlled lab environment. We used the same stimuli as in Study 1.

5.1.2. Respondents and procedures

A sample of 82 females ($M_{age} = 21.85$, $SD = 2.77$) completed an online experiment in a controlled lab environment at a large US university. The same data collection method as in Study 1 was used. The participants were randomly directed to one of the three experimental conditions, after which they completed a questionnaire that contained measures for the dependent variables (i.e., appearance satisfaction, attitude toward the ad, and purchase intentions), the mediating variables (i.e., authenticity of the ad, realism of the model, and meaningfulness of the ad), control variables (i.e., novelty, informational value, attractiveness of the model, how current the ad is, and how substantive the ad is), and age. Finally, the respondents were debriefed and thanked for their participation.

5.1.3. Measures

The same measures used in Study 1 were used for the dependent variables (i.e., respondents' appearance satisfaction, attitude toward the ad, and purchase intentions). Authenticity of the ad was measured by one single item on a 100-point slider scale (i.e., "*How authentic is this ad to you*?"). Meaningfulness of the ad was measured by a single item on a 100-point slider scale (i.e., "*How meaningful is this ad to you*?"). Realism of the model was measured by a single item on a 100-point slider scale (i.e., "*How real is this model to you*?"). Novelty of the ad, informational value, attractiveness of the model, and how current

the ad is were measured each by a single item on a 100-point slider scale (e.g., "*How new is this ad to you*?").

5.2. Results

5.2.1. Manipulation checks

Manipulation checks show that the stimuli are identical on the variables of interest, which helps us exclude any alternative explanations (other than authenticity as the mediator) for our effects. We found that the positive effect of the retouch-free disclaimer (over the retouched disclaimer) cannot be attributed to novelty, as both stimuli (i.e., the retouched and the retouch-free disclaimer) were perceived as equally novel ($M_{retouched} = 28.57$; $M_{retouch\text{-}free} = 39.15$; $t(55) = -1.343$ $p = 0.185$). We also found that the stimuli did not differ in a couple of other key variables, such as informational value ($p = 0.095$), attractiveness of the model ($p = 0.419$), how current the ad is ($p = 0.154$), and substantiveness of the ad ($p = 0.078$).

Our sample also shows a remarkably high average (6.44 on a 7-point scale) of advertising literacy and low standard deviation (SD = 0.876), and there is no significant difference among the conditions ($p = 0.653$).

5.2.2. Mediation analyses with authenticity of the ad

To test H3, we conducted a mediation analysis with the three conditions as the multi-categorical predictor (X), appearance satisfaction, attitudes toward the ad, and purchase intentions as the outcome variables (Ys) and authenticity of the ad as a mediator (M). For each of these mediation analyses we used the version 2.15 of the PROCESS macro (http://www.processmacro.org/index.html) by Andrew Hayes. We found the authenticity of the ad to mediate the effect of our disclaimers on women's appearance satisfaction, controlling for their prior appearance satisfaction (covariate). The results show that the effect of the conditions on authenticity of the ad was significant ($p = 0.000$). Also, the effect of authenticity of the ad on appearance satisfaction was significant ($p = 0.000$). The direct effect of our conditions on appearance satisfaction was not significant when authenticity of the ad was entered as mediator ($p = 0.851$). The Bootstrap analysis indicated a significant indirect effect of the three conditions on appearance satisfaction (95% CI = 0.487 to 1.807) through authenticity of the ad. These results confirm that authenticity of the ad acts as a mediator for the effect of the conditions on women's appearance satisfaction.

For the effect of the conditions on attitude toward the ad, we found authenticity of the ad to be the mediator. The results show that the effect of our conditions on authenticity of the ad was significant ($p = 0.000$). Also, the effect of authenticity of the ad on the attitude toward the ad was significant ($p = 0.000$). The direct effect of our conditions on attitude toward the ad was not significant when authenticity of the ad was entered as mediator ($p = 0.332$). The Bootstrap analysis indicated a significant indirect effect of our conditions on attitude toward the ad (95% CI = 0.068 to 0.235) through authenticity of the ad.

In addition, for purchase intentions, we also established authenticity of the ad as a mediator. The effect of our conditions on authenticity of the ad was significant ($p = 0.000$). Also, the effect of authenticity of the ad on purchase intentions was significant ($p = 0.000$). The direct effect of our conditions on purchase intentions was not significant when authenticity of the ad was entered as the mediator ($p = 0.485$). The Bootstrap analysis indicated a significant indirect effect of our conditions on purchase intentions (95% CI = 0.044 to 0.195) through authenticity of the ad. These results confirm H3.

5.2.3. Mediation analyses with realism of the model

Next, we ran a mediation analysis to test whether realism of the model mediates the effect of our conditions on appearance satisfaction (H4). The results show that the effect of our conditions on realism of the model was significant ($p = 0.0001$). Also, the effect of realism of the model on appearance satisfaction was significant ($p = 0.000$). The direct effect of our conditions on appearance satisfaction was not significant when realism of the model was entered as the mediator ($p = 0.203$). The Bootstrap analysis indicated a significant indirect effect of our conditions on appearance satisfaction (95% CI = 0.005 to 0.061) through realism of the model. These results confirm H4.

5.2.4. Mediation analyses with meaningfulness of the ad

H5 looks into the mediating role of meaningfulness of the ad on advertising effectiveness. For the effect of the disclaimers on attitude toward the ad, we found meaningfulness of the ad to be the mediator. The results show that the effect of our conditions on meaningfulness of the ad was significant ($p = 0.0001$). Also, the effect of meaningfulness of the ad on the attitude toward the ad was significant ($p = 0.000$). The direct effect of our conditions on attitude toward the ad was not significant when meaningfulness of the ad was entered as mediator ($p = 0.489$). The Bootstrap analysis indicated a significant indirect effect of our conditions on attitude toward the ad (95% CI = 0.0005 to 0.0069) through meaningfulness of the ad.

Meaningfulness of the ad also mediated the effect of our conditions on purchase intentions. The effect of our conditions on meaningfulness of the ad was significant ($p = 0.0001$). Also, the effect of meaningfulness of the ad on purchase intentions was significant ($p = 0.000$). The direct effect of our disclaimers on purchase intentions was not significant when meaningfulness of the ad was entered as the mediator ($p = 0.854$). The Bootstrap analysis indicated a significant indirect effect of our conditions on purchase intentions (95% CI = 0.0007 to 0.0079) through meaningfulness of the ad. These results confirm H5.

5.3. Discussion

Study 2 addressed the underlying reason why the condition in which the disclaimer states "This model has not been retouched" has more positive effects on women's appearance satisfaction, attitudes toward the ad, and purchase intentions than the other two conditions (i.e., the disclaimer stating "This model has been retouched" and the control condition without disclaimer). More specifically, Study 2 shows that authenticity of the ad mediates the effect of the three conditions on women's appearance satisfaction, their attitudes toward the ad, and their purchase intentions. In the retouch-free condition, the ad is perceived as more authentic than in the other two conditions, which in turn leads to a higher appearance satisfaction, more positive attitudes toward the ad, and higher purchase intentions.

Literature on authenticity unveils two major components: realism and meaningfulness. Further mediation analyses zooming in on these components show that realism of the model mediates the positive effect of the retouch-free disclaimer on female Millennials' appearance satisfaction. Meaningfulness of the ad mediates the positive effect of the retouch-free disclaimer on female Millennials' attitudes toward the ad and purchase intentions. In sum, Study 2 provides evidence for the role of authenticity and its components in female Millennials' self-evaluation as well as advertising effectiveness.

However, what Study 2 overlooks is the possibility of self-bias when it comes to evaluating physical attractiveness (Wood, 1989). Social comparison theory makes clear that individuals are not always unbiased self-evaluators. They may ignore or distort information as input for social comparison in order to ultimately feed their self-enhancement goals. Hence, following exposure to a model in an ad, it is plausible that they rate their own physical attractiveness differently from how they rate the attractiveness of another, relatable, average-looking person. Study 3 investigates how female Millennials evaluate a non-model woman who is similar to themselves after exposure to our ad disclaimers.

6. Evaluation of other women

Study 1 and Study 2 underline the positive effect of authenticity (a retouch-free campaign) on the evaluation of the self and advertising

effectiveness. Our next study (Study 3) considers the impact of such a campaign on the evaluation of other women, to tackle the possibility of self-bias in a social comparison context. In other words, Study 3 aims to investigate whether women engage in self-bias in order to achieve self-enhancement.

From an intuitive perspective, we would also expect a retouch-free campaign to have a positive impact on how women evaluate the attractiveness of other women. A retouch-free, authentic campaign should increase the amount of attractiveness a woman expects to see in another woman. However, this might not be the case according to social comparison theory. If a retouch-free campaign increases evaluation of the self and other women represent a potential threat in terms of a reference point for one's own level of attractiveness, it is possible that a retouch-free campaign (versus a retouched one) could decrease the perception of attractiveness of other women, in order to ultimately boost self-enhancement. In other words, bias could come into play. When females are exposed to a retouch-free, authentic campaign, they are reminded (according to Study 1) that what is natural and authentic is good. However, being asked to evaluate another woman who is similar and relatable to them, but who is not a model (downward comparison), represents a potential threat to their authentic self in terms of attractiveness. In 1981, Wills defined the concept of downward social comparison as a tendency that is activated when we consider an individual or group to be less of value in order to feel better about ourselves. Competitiveness increases as proximity to a standard of performance increases, and an average-looking woman represents an approachable standard of comparison. Therefore, evaluation of the other woman in terms of attractiveness is expected to decrease with exposure to a retouch-free versus a retouched campaign.

H6. After exposure to the retouch-free disclaimer, females evaluate a non-model woman as less attractive than after exposure to the retouched disclaimer.

7. Study 3

Study 3 – preceded by a pretest – tests the effect of advertising disclaimers regarding digitally enhanced models on women's evaluation of other non-model women.

7.1. Method

7.1.1. Pretest non-model woman

To select a non-model woman, we conducted a pretest with five pictures of young non-model women publicly available online, who were facing the camera in close-ups with a neutral background. The selection was based on the following criteria: Millennials' self-identification with the woman, her similarity to an average-looking woman, and her perceived age. In this pretest, 102 female Millennials (M_{age} = 20.96) recruited from a large southwestern US university saw five pictures of average non-model women in a within-subjects design. They were asked to indicate to what extent they self-identified with the women, to rank them in terms of similarity to an average woman, and to estimate their age. The pretest results showed that one particular woman received the highest score in combination with the lowest standard deviation on self-identification ("To what extent do you identify yourself with this woman?") (M = 2.13, SD = 0.864). When asked to rank order the five women in terms of how "average-looking" they were (with "average-looking" meaning "most neutral, most representative of the average-looking woman"), the respondents ranked her highest (M = 2.36, Mode = 1, with 1 being the most average-looking woman, and 5 being the least average-looking woman). Her age was guessed to be 24.26 on average (SD = 2.61). Appendix B illustrates the picture of the non-model woman we used in Study 3.

7.1.2. Design and stimuli

Study 3 comprises a three-level one factor (advertising disclaimer: "retouched" vs. "not retouched" versus control group containing no disclaimer) between-subjects online experiment in a controlled lab environment. We used the same stimuli as in Study 1 and 2 and the picture of the non-model woman (Appendix B).

7.1.3. Respondents and procedures

A sample of 87 female Millennials (M_{age} = 21.71, SD = 2.06) completed an online experiment in a controlled lab environment at a large US university. The same data collection method as in Study 1 was used. The participants were randomly directed to one of the three experimental conditions, after which they completed measures for age and gender. Next, they were exposed to a picture of a normal (i.e., non-model) woman, after which they were asked to complete the measure for the dependent variable (i.e., evaluation of a non-model woman). Finally, the respondents were debriefed and thanked for their participation.

7.1.4. Measures

Respondents' evaluation of the non-model woman was measured with VAS as adopted by Tiggemann et al. (2014) by using a single item on a 100-point slider scale.

7.2. Results

We found a significant effect between the three conditions on women's evaluation of a non-model woman ($F(2, 86) = 5.60$, $p < 0.01$). Further t-testing shows that – after exposure to the "retouch-free" disclaimer – women (M = 42.42) rated the attractiveness of a non-model woman significantly lower than after exposure to a "retouched" disclaimer (M = 66.81) ($t(47) = 4.18$, $p < 0.001$), which confirms H6. The retouch-free disclaimer and the control condition did not differ significantly from each other ($t(69) = -1.08$, $p = 0.284$). Fig. 4 illustrates these results.

7.3. Discussion

Study 3 demonstrated that the effect of advertising disclaimers on women's evaluation of the attractiveness of other (non-model) women is different from their evaluation of their own attractiveness (Study 2). Specifically, we found that a retouch-free disclaimer, which was found to boost women's own appearance satisfaction (as shown in Study 1), leads to a lower evaluation of other women's physical appearance. This confirms social comparison theory's idea of upward comparison (Study 1) and downward comparison (Study 3). When women are exposed to an advertisement that is labeled as "retouch-free," they use that (supposedly real) model as their benchmark of (upward) comparison and increase their self-evaluation to meet their ultimate goal of

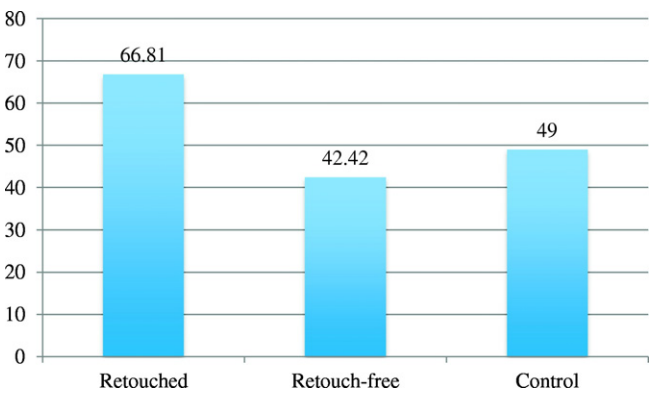


Fig. 4. Study 3 – evaluation non-model woman results.

self-enhancement. The beautiful and real model is part of the "club" they aspire to belong to. However, when women are first exposed to an advertisement that is labeled as "retouch-free" and subsequently asked to evaluate the attractiveness of another, regular woman who is similar to themselves, rather than rate their own attractiveness, this other normal woman becomes the target of comparison. This woman is not part of their aspiration group but rather part of their peer group. This is a situation of downward comparison, and in order to make themselves feel better (i.e., the ultimate goal of self-enhancement), they will rate the other woman as less attractive.

8. General discussion

This research investigates how a "retouch-free" versus a "retouched" disclaimer affects women's appearance satisfaction, attitude toward the ad, and intentions to purchase from the brand, as well as their evaluations of other (non-model) women. In three studies we explained why a "retouch-free" disclaimer is more effective than a "retouched" disclaimer when it comes to self-evaluation and advertising effects, but not with regard to the evaluation of others. Study 1 showed that female Millennials have greater appearance satisfaction, attitude toward the ad, and intentions to purchase the brand when exposed to a disclaimer that states that the model has not been retouched, as opposed to a disclaimer that states that the model has been retouched or an ad without a disclaimer. Study 2 revealed the underlying reasons for the hypothesized effects in Study 1 by identifying authenticity of the ad as the significant mediator of appearance satisfaction, attitude toward the ad, and purchase intentions from the brand. Further, Study 2 also defined the concept of authenticity with greater granularity, by revealing its two major components: realism and meaningfulness. We found that realism of the model mediates the positive effect of a retouch-free disclaimer on women's appearance satisfaction, whereas meaningfulness of the ad mediates the positive effect of a retouch-free disclaimer on women's attitudes and purchase intentions. Finally, Study 3 tackled the possibility of self-bias by investigating how women rated the physical attractiveness of other non-model women following exposure to our advertising disclaimers. Study 3 showed that the retouch-free disclaimer, which had positive effects on women's evaluation of their own attractiveness, does not have the same effect on their evaluation of others; in fact, female Millennials rated other female Millennials they identified with (peer group) as less attractive following exposure to a retouch-free disclaimer. Appendix C provides an overview of all studies and results.

8.1. Theoretical implications

First, our research contributes to the growing body of research on advertising disclaimers by elucidating contradicting prior research about the effectiveness of advertising disclaimers on appearance satisfaction (Tiggemann et al., 2013; Tiggemann et al., 2014; Want, 2009; Slater et al., 2012; Grabe et al., 2008; Bury et al., 2014). While other research found that retouch-free and retouched disclaimers are not perceived differently (Tiggemann et al., 2014), our findings show that a retouch-free disclaimer exerts a more positive effect. A real campaign (retouch-free) activates a beneficial upward comparison where women might rate their attractiveness as closer to women they are compared with. The labeling of images as not digitally enhanced may serve to increase perceived self-evaluation and therefore increase appearance satisfaction.

Second, our research is the first to our knowledge to consider the effect of different disclaimers not only on appearance satisfaction, but also on attitude toward the ad and purchase intentions of the brand.

Third, this research contributes to the literature by establishing authenticity as the mediator of the effects on appearance satisfaction, attitudes, and purchase intentions, while also unraveling the construct of authenticity into its main components: realism and meaningfulness. This study contributes to research on what constitutes authenticity in print advertising by demonstrating that authenticity of the ad drives the positive effects of a retouch-free campaign. Our results are in line with previous studies on advertising and branding by showing that a focus on authenticity by advertisers can have beneficial effects on the evaluation of ads, products, and brands.

Lastly, our research contributes to the literature on self-bias by testing the effects of advertising disclaimers not only on women's self-evaluation but also on their evaluation of other women.

8.2. Managerial implications

Our results call into question advocacy efforts to label retouched media images as a strategy to decrease women's dissatisfaction with their own appearance. Our results have high managerial relevance for both public policy advocates as well as advertisers trying to understand how to persuade consumers based on the concept of authenticity.

For public policy advocates, our results are striking in two ways. First, our results weaken the importance of labeling retouched advertising images, which several countries have heavily advocated in the past few years. Our results show that a retouched disclaimer might not exert the desired effect. In fact, it might not make any difference compared to not having a disclaimer at all, since young adults (women in this case) are knowledgeable about the digital enhancement of images used in advertisement. Second, we found that a retouch-free disclaimer (as opposed to a retouched disclaimer) has a positive effect on female Millennials' appearance satisfaction, attitude toward the ad, and purchase intentions. This is attributed to authenticity: ads that are accompanied by "retouch-free" disclaimers present a more realistic model and a more meaningful message, which is well received by female Millennials. Remarkably, however, this positive effect does not extend to their evaluation of other women's attractiveness. This information is valuable to public policy advocates in that it helps understand female Millennials' reasoning when exposed to different advertising disclaimers and how these affect their mental well-being.

For advertisers, our study underscores that – despite the advertising industry still being characterized by retouched images and messages that appeal to the fantasy of the consumers – Millennials call for authenticity. Our results show that a focus on being retouch-free resonates with consumers more than admitting to being retouched. Consumers do not need to be reminded that "this model has been retouched" because they already know. Instead, Millennial consumers value advertisements that are authentic, containing realistic models, and meaningful messages. Millennials do not just want to hear about how authentic you are; they want you to actually be authentic (Byron, 2016). Our study is the first to our knowledge to identify the key ingredients (i.e., realism of the model and meaningfulness of the ad) that serve as workable tools for practitioners promoting authenticity in print ads.

That being said, it is important for practitioners not to underestimate the power of Millennials. Who are Millennials? According to Byron (2016), there are 75 million in the US population. They are an important segment of empowered consumers who might change the rules of the advertising industry. Given their demand for authenticity, the question for advertisers becomes how to achieve authenticity in the design and delivery of an advertising message. We contribute to answering that question by identifying, explaining, and testing the hypothesized key ingredients of authenticity, realism, and meaningfulness in print advertising.

8.3. Limitations and further research

This study contributes to advertising literacy and represents the first step in the understanding of how marketers play a role in the definition of authenticity and how consumers might relate to it. Our research consisted of experimental studies that were limited to short-term

effects of advertising stimuli. However, our results may help to interpret non-causal research that seeks to investigate long-term cumulative effects of exposure to advertising stimuli. Research on the role of gender in advertising (for a review see Van Hellemont & Van Den Bulck, 2012) suggests that physical attractiveness is still a key element in the advertising of women to women, while men are not usually held to this standard. Nevertheless, future studies should consider the impact of disclaimers on men as well.

Like all studies, our findings should be interpreted in light of some limitations. First, our findings are restricted to the specific stimuli used – that is, one specific advertisement modified based on the presence (two conditions) or absence (control condition) of the disclaimer. Therefore, they may not be applicable to other magazine content or media images (Want, 2009). A second limitation is that our studies did not include a quantitative measure for perceived similarity with the model. Our stimuli were selected based on a qualitative pre-test using the criteria of similarity in terms of ethnicity and age. While the results of our pre-test show that the female Millennials in our target group identified themselves strongly with the model based on these criteria, it would be advisable for further research to include a quantitative measure that would allow for the calculation of a distance type of measure – how far an individual perceives herself to be from the model in the ad. On that note, further research should also take into account changing trends in the beauty ideal or different beauty ideals in different cultures. The model in this research was young, thin, and white. Further research could investigate whether different models would be more appropriate in other cultures, or what the impact would be of older models in ad campaigns targeted at older consumers who already have signs of aging (e.g., wrinkles).

Third, our ads were not presented in the competitive environment of other ads. There may have been more time spent on the stimuli in this study compared to a real scenario (i.e., flipping a magazine). Therefore, future studies should consider the possibility of testing these ads in a competitive environment of other ads. It would enhance the external validity of our results to test the effect of our ad among multiple retouched fashion ads in a print magazine.

Can these "real" campaigns make a difference when young Millennials are still flooded with perfectly retouched images across various media? Testing this effect could also provide insight on the retouch-free condition in relation to social desirability.

Future studies should also explore when our particular effects are likely to be observed and at what point they may even reverse. For example, Millennials want authenticity. However, is this contingent upon certain criteria being met (e.g., model attractiveness)? Would they expect touching up when a model was not meeting their expectations in terms of attractiveness? Future studies should explore when touch ups are preferred and when they are not. Moreover, in many cases the purchaser is not the consumer. A husband may purchase lingerie or perfume for his partner based on an ad. Future studies should test if the results hold in scenarios where the purchaser is not the end user of the product.

Another interesting avenue for further research would be longitudinal: it would be of added value to collect prior measures of self-worth, appearance satisfaction, etc. Then, weeks later, a study could be conducted to assess a possible increase/decrease in scores on those variables. In the present study, we did include these prior measures as control variables, but the time difference between measuring them and the post measures was not long enough to constitute a longitudinal study.

Regarding our mediating variables, further research could also manipulate meaningfulness and realism directly in the ads and then measure their effects on appearance satisfaction, attitudes, purchase intentions, and evaluation of other women's attractiveness.

Future studies could also consider the use of eye-tracking devices. Thanks to eye-tracking technology, we can tell what really attracts consumers' attention when they are looking at an ad. In our particular study, eye-tracking would establish whether the disclaimer impacts women's visual attention to the entire advertisement. Research by Bury et al. (2014) that employed eye-tracking showed that women do attend advertising disclaimers, but this research did not explore the effect of a retouch-free ad vs. one with a retouched disclaimer on the entire advertisement.

Despite these limitations, our study has demonstrated that a retouch-free campaign might exert a more positive impact than a retouched campaign. Being authentic is the key ingredient needed to create authentic print advertising messages that will resonate with consumers in terms of realism and meaningfulness of the ad.

Appendix A. Note: The image was adapted from the Aerie real campaign (source of the image: http://www.thelingerieaddict.com/2014/01/im-excited-new-aerie-real-campaign.html)



Image 1. Retouched advertising disclaimer.



Image 2. Retouch-free disclaimer.

Appendix B



Image 3. Non-model woman used for Study 3.

Appendix C. Overview of all the studies and results

Study	Test	Sample size	Purpose	Dependent variables	Results
1	Pre-test 1	$N = 15$	Selection stimuli	Ethnicity, self-identification, attractiveness, familiarity, credibility	Most adequate stimulus selected (see Appendix A)
	Pre-test 2	$N = 126$	Pre-test stimuli	Quality	$p = 0.319$
				Believability	$p = 0.601$
				Informational value	$p = 0.09$
				Professionalism	$p = 0.241$
			Check advertising literacy	Advertising literacy	M = 5.28, SD = 0.84 (7-point scale)
	Main study	$N = 230$	Describe sample characteristics	Prior appearance satisfaction	M = 68.21, SD = 21.38 (100-point scale)
				Prior self-esteem	M = 3.33, SD = 0.49 (4-point scale)
				Prior mood	M = 57.95, SD = 22.46 (100-point scale)
			Sample control across 3 conditions	Age	$p = 0.367$
				Prior mood	$p = 0.825$
				Self-esteem	$p = 0.713$
			Test H1	Appearance satisfaction	$F(2, 229) = 3.117$, $p = 0.046$ $M_{retouch\text{-}free} = 58.94$; $M_{retouched} = 48.82$; $t(139) = -2.367$, $p = 0.019$
			Test H2	Attitudes toward the ad	$F(2, 227) = 5.83$, $p = 0.003$ $M_{retouch\text{-}free} = 4.19$; $M_{retouched} = 3.55$; $t(139) = -2.955$, $p = 0.004$
				Purchase intentions	$F(2, 243) = 7.18$, $p = 0.001$ $M_{retouch\text{-}free} = 3.30$; $M_{retouched} = 2.75$; $t(148) = -2.071$, $p = 0.040$
2	Main study	$N = 82$	Manipulation check	Novelty of the ad	$p = 0.185$
				Informational value of the ad	$p = 0.095$
				Attractiveness of the model	$p = 0.419$
				Currency of the ad	$p = 0.154$
				Substantiveness of the ad	$p = 0.078$
			Check advertising literacy	Advertising literacy	M = 6.44, SD = 0.876 (7-point scale)
			Test H3 (authenticity as mediator)	Appearance satisfaction	The Bootstrap analysis: indirect effect of conditions on appearance satisfaction (95% CI = 0.487 to 1.807) through authenticity of the ad.
				Attitudes toward the ad	The Bootstrap analysis: indirect effect of conditions on attitude toward the ad (95% CI = 0.068 to 0.235) through authenticity of the ad.
				Purchase intentions	The Bootstrap analysis: indirect effect of conditions on purchase intentions (95% CI = 0.044 to 0.195) through authenticity of the ad.
			Test H4 (realism of the model as mediator)	Appearance satisfaction	The Bootstrap analysis: indirect effect of conditions on appearance satisfaction (95% CI = 0.005 to 0.061) through realism of the model.
			Test H5 (meaningfulness of the ad as mediator)	Attitudes toward the ad	The Bootstrap analysis: indirect effect of conditions on attitude toward the ad (95% CI = 0.0005 to 0.0069) through meaningfulness of the ad.
				Purchase intentions	The Bootstrap analysis: indirect effect of conditions on purchase intentions (95% CI = 0.0007 to 0.0079) through meaningfulness of the ad.
3	Quantitative pre-test	$N = 102$	Selection of non-model woman	Self-identification	M = 2.13, SD = 0.864
				Most average-looking	M = 2.36, Mode = 1, with 1 = most average looking woman
				Estimated age	M = 24.26, SD = 2.61
					Most adequate picture selected (see Appendix B)
	Main study	$N = 87$	Test H6	Attractiveness evaluation of other non-model woman	$F(2, 86) = 5.60$, $p < 0.01$ $M_{retouch\text{-}free} = 42.42$; $M_{retouched} = 66.81$; $t(47) = 4.18$, $p < 0.001$

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